CareNexus

4-Week Team Development Roadmap

Smart Hospital Service Comparison & Navigation Platform
A strict dependency-based roadmap for a 4-member team

Project Development Principle

This roadmap is designed so that work flows in sequence. Each member must complete and hand over the required output before the next dependent member starts that task. This structure follows the same staged roadmap style used in the uploaded AI CHAT BOT USING RAG roadmap, where weekly features lead to a defined milestone. ■filecite■turn0file0■L1-L18■

Core CareNexus logic: Required Service → Confirm Availability → Find Suitable Hospitals → Compare/Rank → Select Hospital → Route & Navigation.

Overall Dependency Flow

Member 1 → Member 2 → Member 3 → Member 4
No dependent task should move forward until the previous handover is completed.

WEEK 1: Hospital Data & Service Foundation

Goal: Build the structured foundation required for every later feature: hospital information, service definitions, service availability and database storage.

Order	Member	Assigned Feature	Technical Focus	Required Handover
1	Member 1	Hospital Data Model	Define hospital fields: ID, name, address, city, latitude, longitude, contact and rating.	Hospital data schema and sample data are ready.
2	Member 2	Service Catalog	Create standardized service records such as Emergency, ICU, MRI, Cardiology, Orthopedics and Blood Bank.	Service catalog linked to the hospital data model.
3	Member 3	Service Availability Mapping	Create the hospital-to-service relationship with statuses such as Available, Limited, Unavailable and Unknown.	Validated hospital-service availability mapping.
4	Member 4	Database Integration	Create PostgreSQL tables and relationships. Insert and validate the combined data.	Working database containing hospitals, services and availability.

Strict dependency: Member 1 schema → Member 2 service catalog → Member 3 availability mapping → Member 4 database integration.

End of Week 1 Milestone: A unified hospital database where the system can answer: “Does this hospital provide this service?”

Languages/Tech: Python, SQL, PostgreSQL, SQLAlchemy.

WEEK 2: Backend Search, Filtering & Comparison

Goal: Turn the Week 1 data into working backend functionality. Service availability must be checked before comparison.

Order	Member	Assigned Feature	Technical Focus	Required Handover
1	Member 1	FastAPI Foundation	Create FastAPI project, database connection and base hospital/service endpoints.	Running API and documented base endpoints.
2	Member 2	Hospital Search API	Build search using hospital name, location and available database fields.	Search API returning structured hospital results.
3	Member 3	Service Availability Filter	Filter search results so only hospitals matching requested services are returned.	Filtered results API ready for comparison.
4	Member 4	Comparison Engine	Compare 2–4 filtered hospitals using service match, facilities, distance-related data and other stored criteria.	Comparison response/API ready for frontend.

Strict dependency: API foundation → hospital search → availability filtering → hospital comparison.

End of Week 2 Milestone: A backend user can request services such as “MRI + Emergency”, receive only matching hospitals, and compare selected results.

Languages/Tech: Python, FastAPI, SQL, PostgreSQL, SQLAlchemy.

WEEK 3: Location, Recommendation & Route

Goal: Add location intelligence and navigation. The route is generated only after suitable hospitals have been identified.

Order	Member	Assigned Feature	Technical Focus	Required Handover
1	Member 1	Location Foundation	Use user location input/permission and hospital latitude/longitude from the database.	Location data interface and coordinates available.
2	Member 2	Nearby Hospital Logic	Calculate/filter nearby hospitals from the service-matched hospital set.	Suitable hospitals enriched with distance information.
3	Member 3	Smart Ranking	Rank hospitals using service match as highest priority, then distance and other available criteria.	Ordered recommendations with score/reason data.
4	Member 4	Route & Navigation	Integrate maps/routing provider for the selected recommended hospital.	Route, distance and directions flow working.

Strict dependency: Get location → calculate nearby suitable hospitals → rank hospitals → generate route to selected hospital.

Important rule: CareNexus should not recommend a hospital solely because it is nearest if the requested service is unavailable.

End of Week 3 Milestone: The system identifies hospitals with the required service, ranks suitable choices and provides navigation to the selected hospital.

Languages/Tech: Python, JavaScript/TypeScript, Maps/Routing API.

WEEK 4: Frontend, End-to-End Integration & Testing

Goal: Build the complete user experience and connect all previously completed modules.

Order	Member	Assigned Feature	Technical Focus	Required Handover
1	Member 1	Search Interface	Build service selection, hospital search and location input UI.	Search UI sends correct requests.
2	Member 2	Hospital Results UI	Display service-matched hospitals returned by the backend.	Users can view and select matching hospitals.
3	Member 3	Comparison & Recommendation UI	Show side-by-side comparison and ranked recommendations.	Comparison and recommendation screens working.
4	Member 4	Map, API Integration & Testing	Connect all frontend actions to APIs, maps/routes, loading states and errors; perform end-to-end testing.	Complete integrated CareNexus application.

Strict dependency: Search UI → results UI → comparison/recommendation UI → API + map integration + final testing.

End of Week 4 Milestone: A complete working application: Search Service → Verify Availability → Display Matching Hospitals → Compare → Recommend → Select → Navigate.

Languages/Tech: HTML, CSS, JavaScript/TypeScript, React, Python/FastAPI APIs.

Complete Member-Wise Dependency Summary

Member	Primary Responsibility	Depends On	Output Used By
Member 1	Foundation of each weekly pipeline	Previous week's completed milestone	Member 2
Member 2	Search/data processing stage	Member 1 handover	Member 3
Member 3	Filtering/ranking/comparison stage	Member 2 handover	Member 4
Member 4	Integration/final stage	Member 3 handover	Next week's full team milestone

Recommended Tech Stack

Layer	Technology	Language
Frontend	React	JavaScript or TypeScript
Styling	CSS / Tailwind CSS	CSS
Backend	FastAPI	Python
Database	PostgreSQL	SQL
ORM	SQLAlchemy	Python
Maps & Routing	Maps provider/API	JavaScript/TypeScript + API
Version Control	Git + GitHub	Git
Testing	Pytest + frontend testing tools	Python + JavaScript/TypeScript

Final Project Flow

User Location / Search → Required Medical Service → Service Availability Verification → Suitable Hospitals → Distance Calculation → Comparison & Ranking → Hospital Selection → Route & Navigation

Final Project Title

CareNexus: An Intelligent Hospital Service Comparison and Navigation Platform

Find the Right Service. Compare the Right Hospitals. Choose the Right Route.